

Flu Vaccination Predictor Project

By ALPHA GROUP

Assimagbe Albert Raphael & Saad Ali



Based on CDC H1N1 2009 Survey
Data

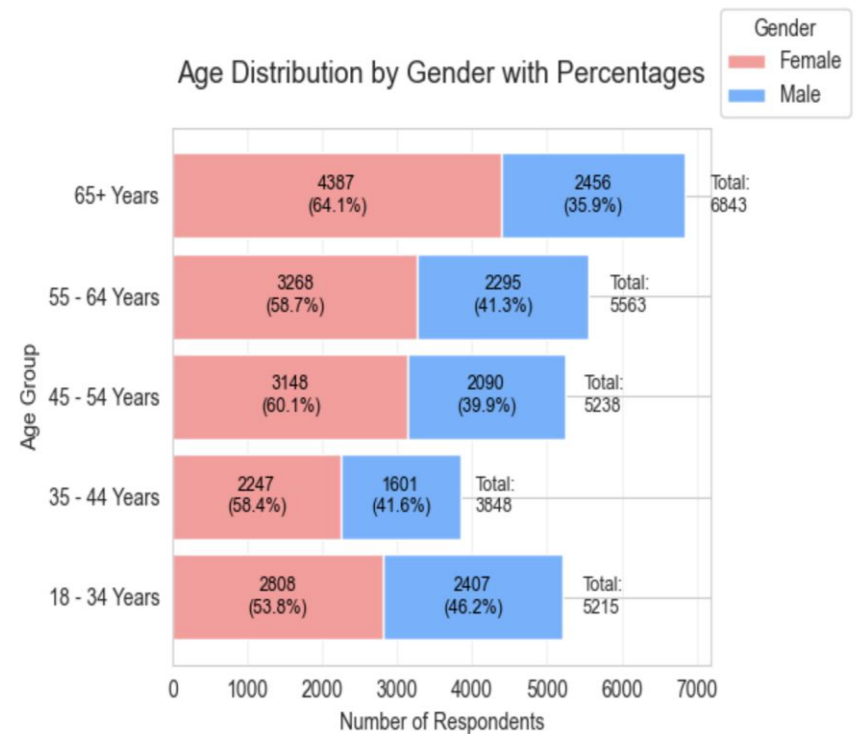
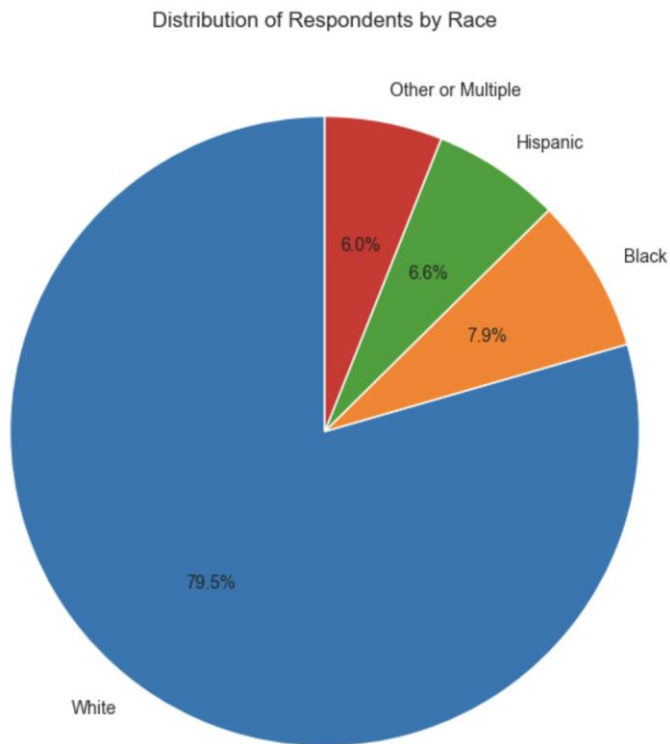
Project Goals & Questions

- **Objectives:**
 - - Identify who gets flu vaccines and why
 - - Understand role of doctor recommendations
 - - Use machine learning to predict vaccine behavior
 - - Suggest smart ways to improve vaccination
- **Key Questions:**
 - - Who avoids the vaccine?
 - - Do doctor suggestions help?
 - - Which behaviors matter most?

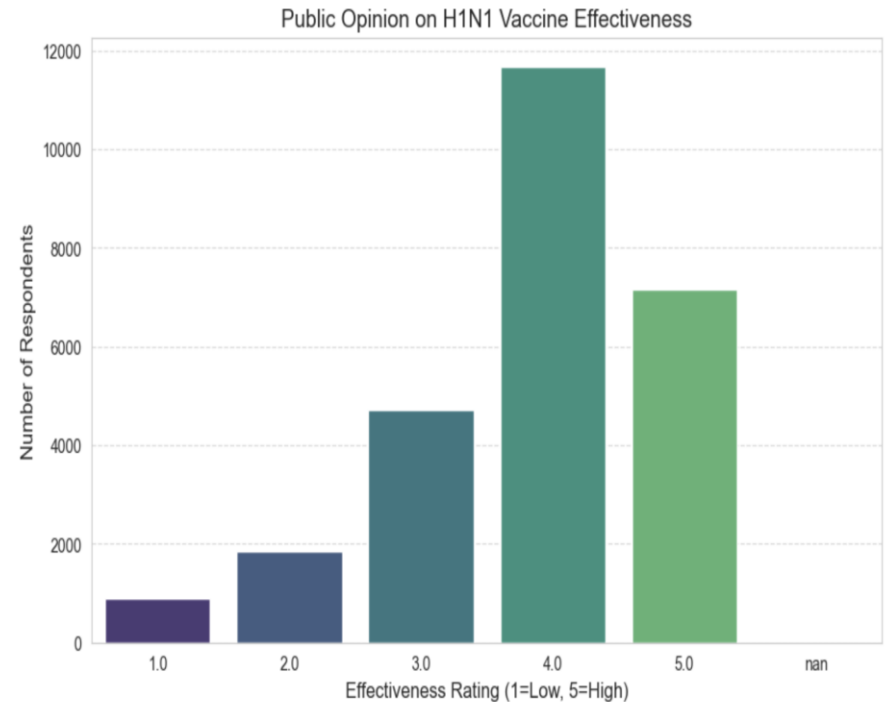
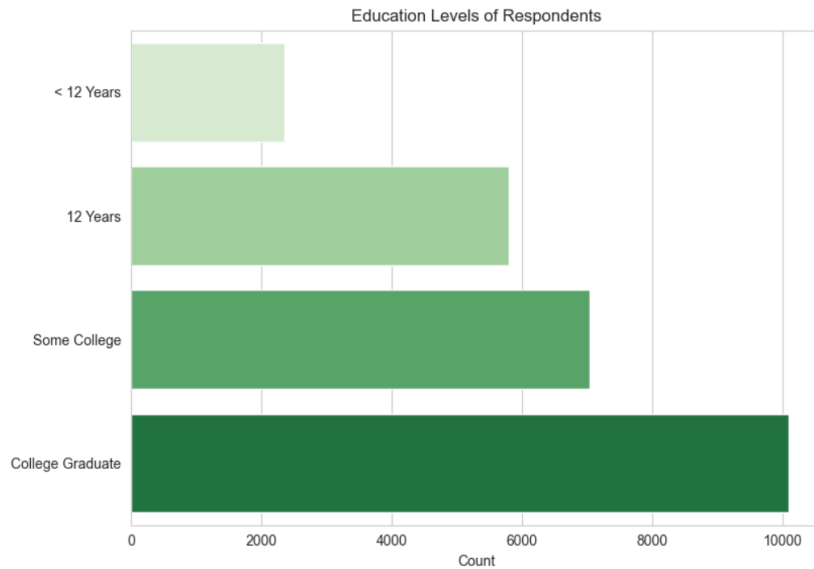
Dataset Summary

- Source: CDC 2009 H1N1 Survey
- Size: 26,707 people, 36 features
- **Data Includes:**
 - - Demographics (age, education, race, gender, etc.)
 - - Behaviors (handwashing, avoidance)
 - - Health opinions and access
- **Challenge:** Missing data in some columns (up to 50%)

Visuals Of Race & Age Distributions

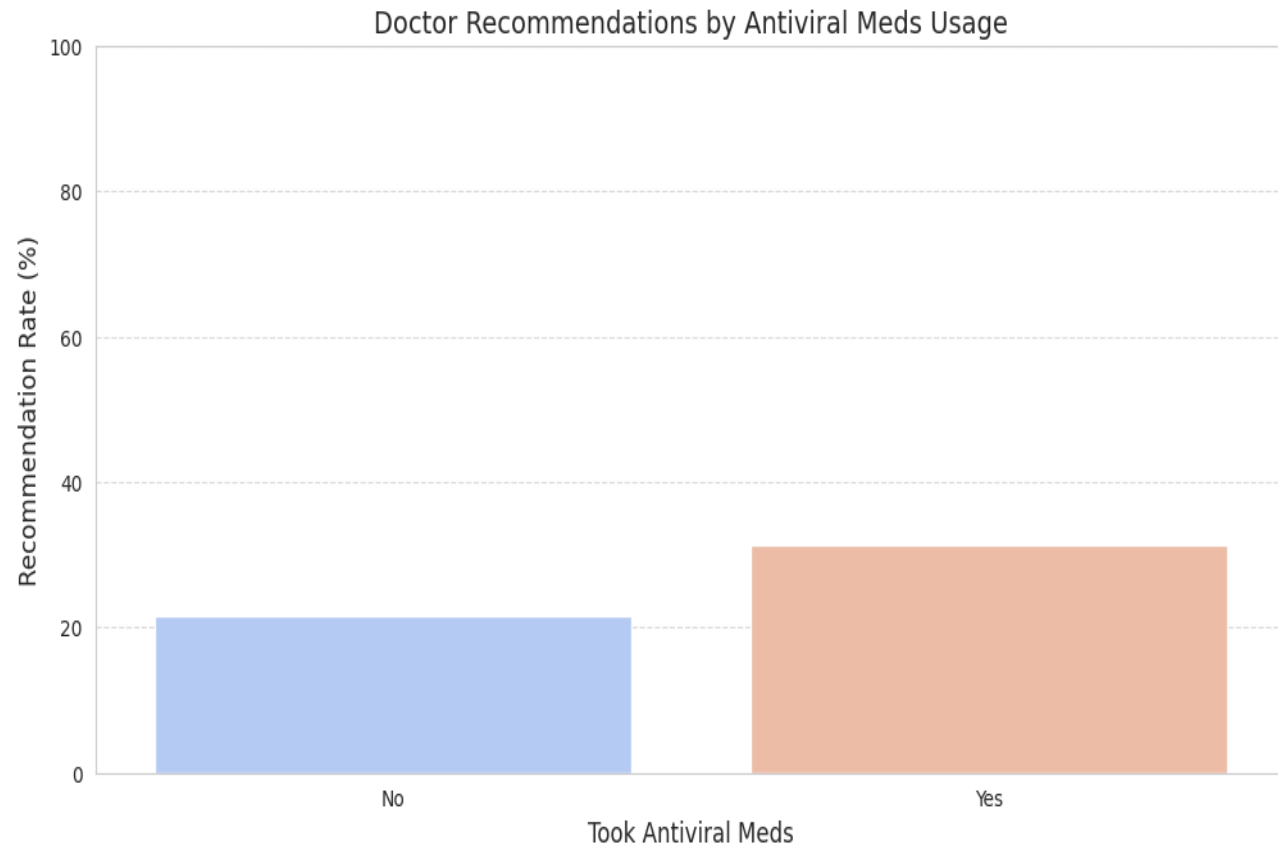


Visuals Of Educational Levels & Public Opinion



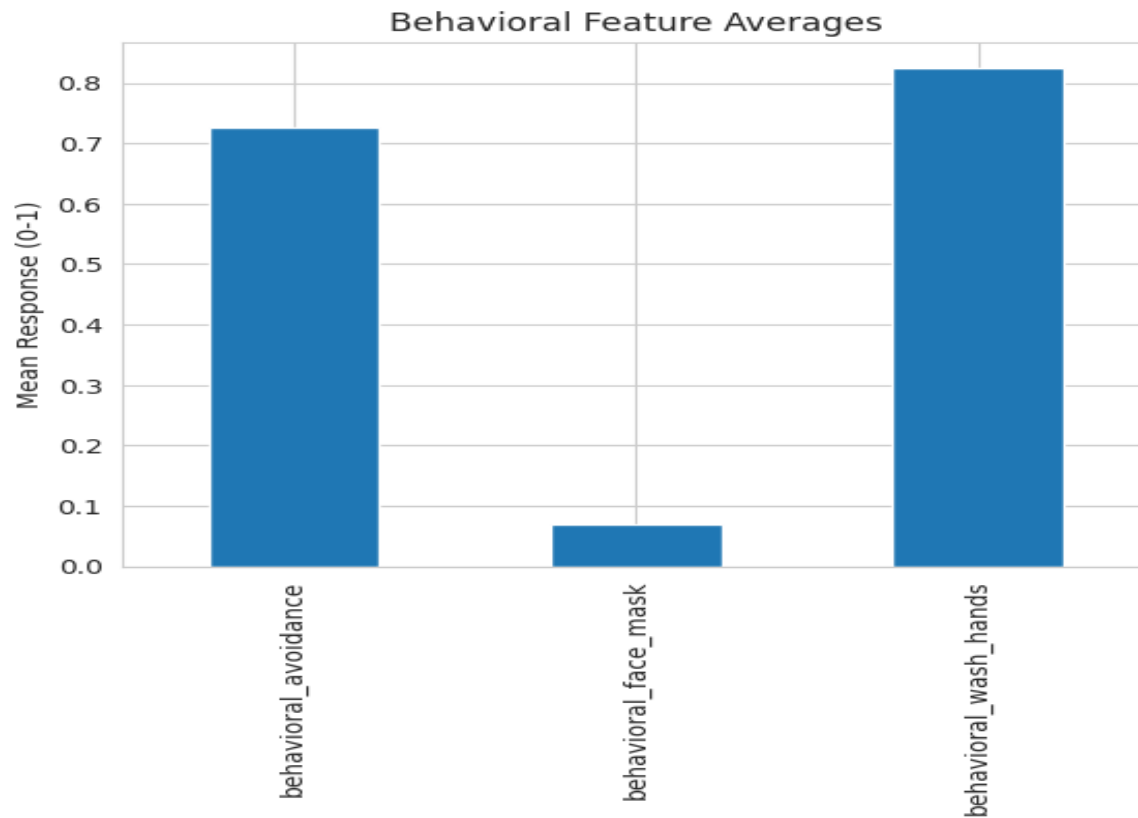
Key Findings & Visuals

1. Doctor Recommendations Influence Vaccination



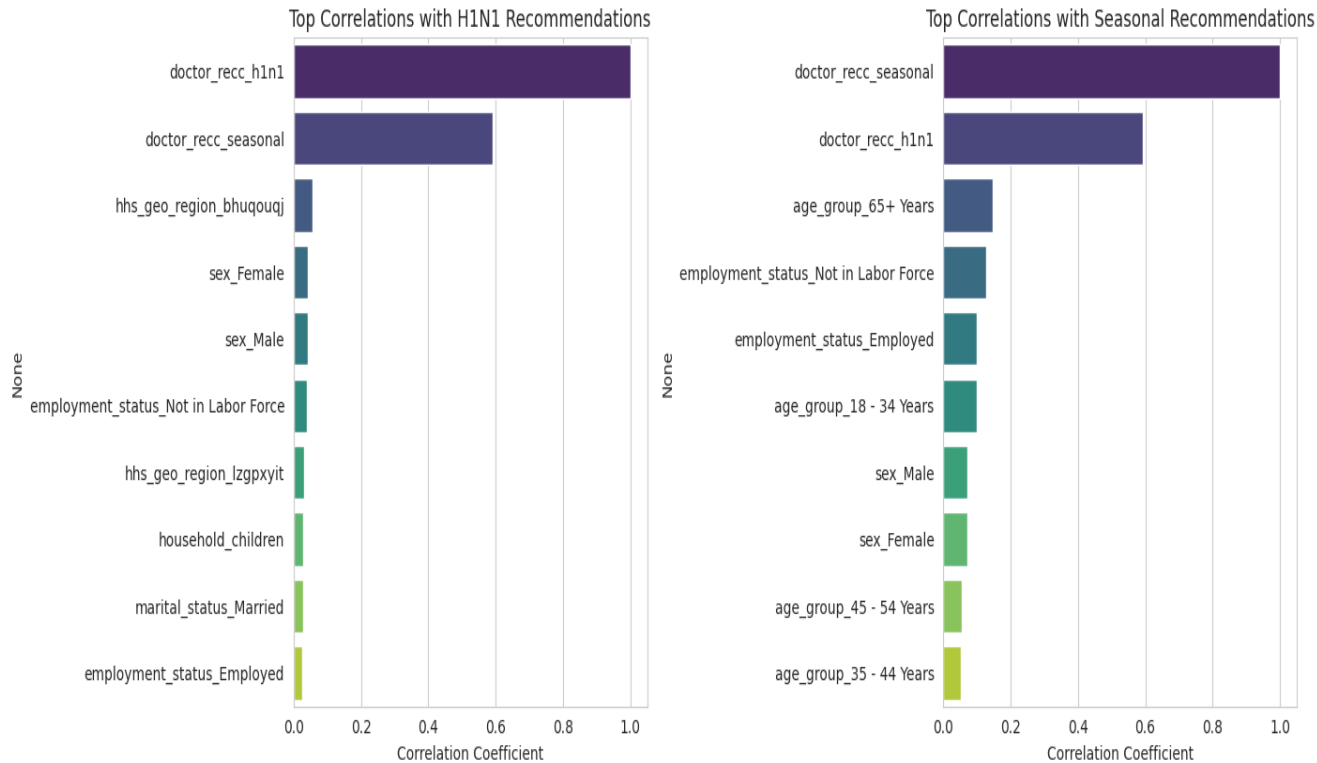
Key Of Findings

2. Concern about H1N1 Virus = more handwashing



Key Findings & Visuals

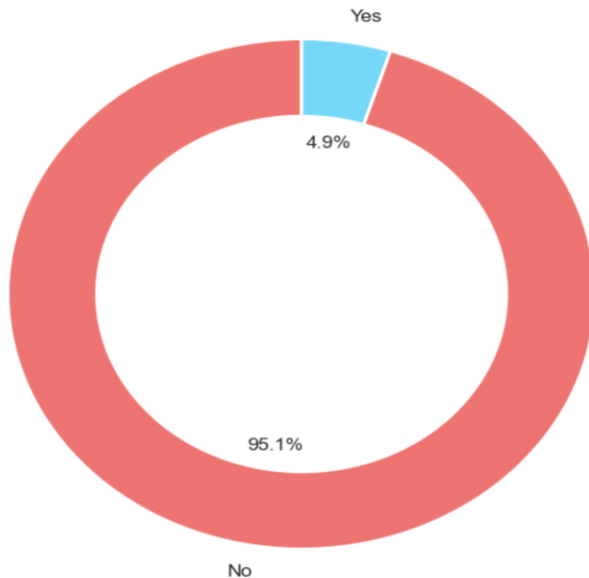
3. Older adults (65+) more likely to get seasonal flu shots



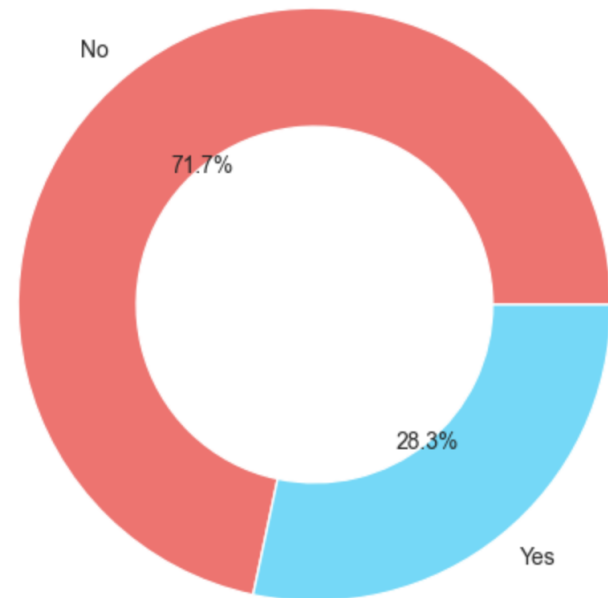
Key Findings & Visuals

4. 4.9% Persons takes Preventive Antiviral Medication while, 28.3% Persons has chronic medical conditions

Preventive Antiviral Medication Usage



Chronic Medical Conditions



Model & Analysis

- - Cleaned all missing data (used imputation or dropped high-missing columns)
- - Used logistic regression and feature importance to analyze impact
- **Important Analysis:**
 - - Doctor recommendation (47.5%)
 - - H1N1 Virus concern (57.2%)
 - - Vaccine effectiveness belief
 - - Preventive Antiviral Medical Usage (4.9%)
 - - Chronic conditions (28.3%)

Recommendations

1. Train doctors to give strong, clear vaccine advice
2. Target under-vaccinated groups (50–64 year-olds)
3. Use SMS campaigns and infographics
4. Promote combined vaccine confidence messages
5. Improve future surveys: ask about access and timing

Final Thoughts

- ✓ Machine learning can help target interventions
- ✓ Beliefs and doctor input matter more than demographics
- 🔊 Next step: Build prediction tool for clinics